Completion of the critical six-point-head branch

Human Compression project: L044 continuation

Let $G \leq \mathrm{Alt}(\Omega)$ be transitive with connected string C-marking $\rho_0, \ldots, \rho_{r-1}$, all maximal omissions intransitive, and $n = |\Omega| \geq 12$. Assume the alternating rank bound at every strictly smaller degree at least twelve. All complete acquired proper primitive, anchored imprimitive and finite boundary proofs remain retained. Suppose the first three actual marks form an extendable supported six-point head from source Table 2(2) or (3).

Theorem 1. *One has* $r \leq \lfloor (n-1)/2 \rfloor$.

Proof. Suppose the bound fails. The complete critical-head reduction gives $n = 2r \geq 18$, the literal supported head on D, its actual natural alternating tail $G_{\geq 3}$, and all head and prefix interfaces used below. The full transitive-B exclusion makes $B = G_{\geq 4}$ intransitive on $\Omega \setminus D$. Thus colour 3 is paired; the exact early-cut calculation also makes colour 4 paired. The complete critical-tail theorem supplies a unique ambient-eligible colour at least five. Choose the least such colour j. Every colour $3, \ldots, j-1$ is therefore paired in the full ambient sense.

The pure-prefix unique-cut lemma gives the two actual G_j orbits $C = C_{j-1}$ and $A = \Omega \setminus C$, where C_k is the orbit of $P_k = G_{\leq k}$ containing D. Every mark after j fixes C pointwise. Any shared label lies before j, and its first index g, if present, is at least three.

If the cut is unshared, its two groups are the actual supported transitive prefix and suffix. The complete later-unshared theorem leaves only an actual opposite six-point head, or $j = r-1$ with $A = \{z\}$ and $P_{r-2} = \mathrm{Alt}(\Omega \setminus \{z\})$. The first case is excluded by the complete opposite-head theorem. In the second, minimality gives paired colours $3, \ldots, r-2$. The complete terminal-singleton reduction applies with these exact hypotheses and produces an actual opposite head, again impossible.

It remains that a first shared label g exists. The complete interior and endpoint first-shared-head proofs exclude $g = 3$, using their respective, distinct support interfaces. The complete first-shared-four proof excludes $g = 4$. Hence $g \geq 5$.

Suppose P_{g-1} is transitive on its support. Reverse the entire string and apply the complete transitive-tail alternative at the unique cut. In that reversed orientation the crossing is $i = r-1-j$, its last shared label is $h = r-1-g < r-1$, and its supported transitive tail is exactly P_{g-1}. If $i > 0$, the preceding group is the original $G_{>j}$, which fixes C pointwise by the unique-cut lemma. Thus the chosen-orientation hypothesis is satisfied literally. The exact index-location lemma says that its first-head exit has $g = 3$, already excluded, while its other exit is an actual opposite head with $g = r-3$. The opposite-head theorem excludes that alternative too.

We may therefore assume P_{g-1} is intransitive on its support. If $j = r-1$, the complete endpoint equality route applies: B is intransitive, j is the least later unique colour, and $g \geq 5$. It proves $g = r-3$ and an actual pure opposite Table 2(2) head, already excluded.

For $j < r-1$, the complete first-later-cut equality theorem has all its hypotheses. It gives

$$j = g+1, \quad t = r-g-1 \geq 2, \quad |A| = 2t, \quad \mathrm{Fix}(P_{g-1}) = A \cup \{b\},$$

where $b \in C$ is the crossing endpoint. The support of P_{g-1} has exactly two nontrivial orbits, the head orbit and one exterior orbit, of total size $2g+1$. The suffix $Q = G_{>j}$ has rank $t-1$ and is either transitive on A or has two faithful natural S_t orbits. The complete transitive-equality-suffix theorem excludes the first possibility.

In the second possibility $u = \rho_g$ exchanges the two Q-orbits equivariantly: the earlier prefix fixes A pointwise and $\langle u, Q\rangle$ is transitive on that actual omission orbit. The complete pair-path argument gives the literal marking

$$q_i = (a_{i-1}\ a_i)(a'_{i-1}\ a'_i), \qquad 1 \le i < t, \qquad u|_A = \prod_{i=0}^{t-1}(a_i\ a'_i).$$

For precision, every pair-image omission is intransitive, since adjoining the actual u would otherwise make A one orbit without its omitted, supported mark. One selected edge per label is a tree on t pairs; its omission cuts force each image to be precisely its own edge transposition, and string commutation makes the tree a path. The endpoint anchor fixes its orientation. This argument includes $t = 2, 3$ and does not import an intersection property for a quotient.

The complete small-equality-suffix lemma proves that the crossing $s = \rho_{g+1}$ fixes the old prefix support pointwise, so $s = (b\ a_0)w$ with w an odd involution on $A \setminus \{a_0\}$. If $t \ge 4$, the complete diagonal suffix theorem forces s to fix that entire latter set; then s is a single transposition, impossible in G. If $t = 3$, the complete six-point suffix theorem makes the last ambient omission transitive, also impossible.

For $t = 2$, the complete four-point reduction leaves its one literal decorated form. The complete quotient criterion and its application exclude it: they give a genuine string C-marking of rank $g + 1$ on the natural alternating group of degree $2g + 2 < n = 2g + 6$. Here $g \ge 5$, so the explicit smaller-degree induction applies and gives rank at most g. The retained proof checks the complete quotient kernel, full IP and every nonidentity projected mark; no quotient assertion is inferred from an abstract group identification.

Every least later cut and every actual prefix/suffix orbit case has now been exhausted. The contrary rank range is impossible, proving the theorem. □

Attribution and boundary. This assembles the exceptional-head route underlying Propositions 5.28–5.32 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1. The source cut strategy and all complete acquired replacement proofs remain paid. The assembly uses the general equality route; the additional separately proved exclusions of individual later first-shared indices are unnecessary premises here. This proves the critical head branch, with its induction explicit. The full main-theorem assembly must still account for primitive and imprimitive omissions, the all-intransitive branches without an exceptional head, finite bases and the lower-bound construction. No whole-original, question, numerical or formal credit is claimed.